

YEDUKRISHNAN A V

LEAD MACHINE LEARNING ENGINEER

Adithi Pearl Apartment, NRI Layout,
Bengaluru, Karnataka, India 560016

+91-9829611190 · yeduav@gmail.com · [linkedin.com/in/yeduav](https://www.linkedin.com/in/yeduav)

Lead Machine Learning Engineer with 5 years of active R&D experience in Machine Learning and Deep Learning. Specialized in Natural Language Processing using Deep Learning. Skilled at developing NLP based solutions that optimize compute and cost efficiency while ensuring high levels of accuracy, reliability, scalability and security. Adept at researching and evaluating new ML technologies and deploying them at scale.

PROFESSIONAL EXPERIENCE

THOUGHTFOCUS, BENGALURU – LEAD MACHINE LEARNING ENGINEER

SEP 2019 – PRESENT

- Implemented a completely automated training and inference pipeline for sentiment analysis of student data using BERT. Deployed using AWS Sagemaker Pipeline.
- Document key information extraction framework for extracting key value pairs from given documents.
- Implemented a GenAI based review summarization engine.
- Build and deployed a Semantic Search Engine for improving the retrieval performance of the company's chatbot, YANA.
- Conducted Machine Learning Workshop to upskill 70+ students on Applied NLP in Industry.

Key Achievements : Document key information extraction framework (DocPro) reduced the dependency on readily available cloud frameworks, reduced cost and enabled competitive pricing.

SPOONSHOT, BENGALURU – DATA SCIENTIST

APR 2019 – MAY 2019

- Built a mathematical scoring model to rank ingredient pair combinations using PySpark, Kafka and AWS Sagemaker.
- Wrangled 150-200 TiB of Common Crawl text data to find food related articles using PySpark.

Key Achievements : Improved the ingredient pair scoring mechanism which is the core feature of the Ingredient Recommendation Engine, Genesis.

SPOONSHOT, BENGALURU – DATA SCIENCE INTERN

OCT 2018 – MAR 2019

- Implemented a BERT based classifier for Dish Type prediction, given the details of the dish.
- Built an End to End data scraping pipeline for scraping food related data.

Key Achievements : Reduced the data extraction efforts significantly.

PROJECTS

DOCUMENT KEY INFORMATION EXTRACTION PIPELINE – THOUGHTFOCUS

- Developed a custom NER pipeline for extracting key information from documents.
- Finetuned BERT model for key information extraction.
- Implemented an LLM based data augmentation pipeline to reduce human effort.
- Achieved 80% reduction in document processing time.

RAG PIPELINE FOR YANA CHATBOT – THOUGHTFOCUS

- Used Sentence-BERT as sentence embedding and ChromaDB as the vector store.
- Domain tuned the Sentence-BERT embeddings on custom data to improve retrieval performance.
- Deployed Opensource LLMs at scale using TensorRT in AWS.

AI GENERATED PRODUCT REVIEW SUMMARIES – THOUGHTFOCUS

- Developed a Gen AI based product review summarization tool to generate short summary of the product given verified reviews of the product.
- Data augmentation pipeline for fine tuning the summarization model was implemented to reduce the inference cost and improve performance.

INTERPRETABLE STUDENT DROPOUT PREDICTION AND INTERVENTION – THOUGHTFOCUS

- Developed an XGBoost based model to predict student dropout using academic data.
- Used Interpretable ML and Explainable AI tools to explain individual predictions and enable personalized intervention.

TECHNICAL SKILLS

Programming Languages	:	Python, Java, C/C++, HTML, CSS
Frameworks	:	Pandas, Pytorch, Tensorflow, Numpy, Flask, FastAPI
Data Engineering	:	PySpark, SQL, Kafka, NoSQL, MongoDB
ML	:	Supervised/Unsupervised Learning, Data Analysis
NLP	:	Spacy, Gensim, AllenNLP, Sentence Transformers, Huggingface, Word2Vec, fastText , Langchain
Cloud	:	AWS, Azure
Quick Prototyping	:	Gradio, Streamlit, Langchain
Visualization	:	Weights and Biases, Matplotlib, Seaborn
Orchestration	:	Airflow, MLflow
Optimization	:	LORA

ACADEMIC DETAILS

M.E IN COMPUTER SCIENCE – BITS PILANI, PILANI CAMPUS

2016 – 2018

RELEVANT COURSES : Advanced NLP, Information Retrieval, Advanced Data Mining.

B.TECH IN COMPUTER SCIENCE – RSET, COCHIN

2011 – 2015

RELEVANT COURSES : Computer Vision, Data Mining, Natural Language Processing.

AWARDS AND RECOGNITION

Star Performer in Tech – ThoughtFocus

Awarded for contributions towards Semantic Search related NLP work.

Technology Champ – ThoughtFocus

Awarded for innovation in NLP space across different projects.